

Paper Manager pack — PaliGemma (Sep 16)

Paper: PaliGemma: A versatile 3B VLM for transfer

arXiv: <https://arxiv.org/abs/2407.07726>

PDF: /workspace/papers/paligemma-2407.07726.pdf

Code: https://github.com/google-research/big_vision

Session: Sep 16 · due 9:30 AM · **You = Scientific Reviewer (seat 1)**

Type: VLA Architectures

Sources: Summarizer (/workspace/papers/paligemma-2407.07726-summary.md) + Senior Research (/workspace/paligemma-quiet/DELIVERABLE.md)

1. Paper summary

Open **base** (not instruction-tuned) ~3B VLM for transfer: **SigLIP-So400m** vision + linear connector + **Gemma-2B**. Prefix-LM masking (full attn over image+prefix; causal suffix).

Stages: Stage0=pretrained SigLIP+Gemma → Stage1 multimodal pretrain (~1B examples @224) → Stage2 higher-res (+50M@448, +10M@896) → Stage3 per-task FT.

Headline transfers: COCOcap ~141.9/144.6; VQAv2 ~83.2/85.6; DocVQA **43.7** → **84.8** with resolution; ScienceQA ~95%@224.

Club relevance: Default open substrate for π_0 / $\pi_{0.5}$ / **OpenPI**; BEHAVIOR'25 1st place stack = SigLIP → PaliGemma → Action Expert (~26% q).

2. Keywords

PaliGemma, SigLIP-So400m, Gemma-2B, Prefix-LM, open base VLM, transfer, loc/seg tokens, resolution 224/448/896, VLA backbone, $\pi_0/\pi_{0.5}$ /OpenPI, BEHAVIOR

3. Role packs

Stakeholder (18+3) — seat 4

Commodity open 3B VLM for robot FMs. Prefer paligemma-pt-{224,448}; budget Stage3-like FT; multi-cam often stays @224. Risks: private Stage1 data; license/gating; base≠chat.

Scientific Reviewer (8+3) — ****YOU (seat 1)**** — HIGH PRIORITY

Claim under review: A <3B open *base* VLM transfers broadly and is a better research object than chat-tuned VLMs.

Strengths

- Clear **base vs IT** separation
- Coherent PaLI-line scaling thesis (SigLIP+Gemma)
- Unusually deep ablations (Stage1 length, mask/loss, freeze/reset, connector, encoder, resolution, few-shot)
- Contamination hygiene; honest single-HP transfer protocol
- Structured loc/seg API; systems transparency

Weaknesses

- **Private Stage1 mixture** blocks independent recipe reproduction at paper scale
- Table1 mostly self-vs-resolution, not matched public bake-off
- Metric mixing / macro-average can obscure per-family performance
- Stage3 ≠ OOD robotics transfer
- Square resize may hurt embodiment cameras
- No action / memory / 3D objectives; uneven seeds
- TPU-centric; no GPU/LoRA memory tables in paper (LoRA = Empiricist extension)

Actionable improvements (say these aloud)

1. Public Stage1 surrogate mixture / sampling weights
2. Fixed public baseline panel (Idefics2 / QwenVL / InternVL) under **same** transfer recipe

3. Per-family normalized scores instead of opaque macro-average
4. Robotics probes: egocentric VQA, affordance pointing, multi-cam correspondence
5. Aspect-ratio / camera-model ablations
6. loc/seg calibration + constrained decoding analysis
7. Continuous ViT:LLM LR-ratio sweeps (not binary freeze)
8. Preregister resolution-sensitive tasks; corrected win rates
9. One-command public Table1 harness from released ckpts
10. Negative-results appendix (e.g. SciCap multitask collapse)

Score intuition: Strong open-model + ablation contribution; landmark **if** data opacity is acknowledged; methods venues should demand public mixture + matched baselines.

Empiricist (11+3) — seat 3

- H1: 448pt + LoRA r=16 closes ≥70% of 224 → 448 OCR/VQA gain at <30% full-FT cost
- H2: Native Stage2-res ckpt beats cross-res FT under LoRA
- H3: K-frame concat (4–8 @224) helps short-horizon BEHAVIOR-style VQA vs single frame

Priority	Experiment
P0 E1	Resolution ladder 224/448/(896) on OCR-ish
P0 E2	Freeze matrix (TT / TF / LoRA-LLM / LoRA+connector)
P0 E3	Default-HP vs paper recipe regret
P1	LoRA rank; cross-res transfer; few-shot curves
P2	RefCOCO stretch; multi-frame; encoder-free smoke

Budget: LoRA@224 on ~24GB; full FT @448–896 needs 80GB+. Outlines: /workspace/paligemma-quiet/env_outline/.

Archaeologist (11+3) — seat 2

- Priors: PaLI/PaLI-X/PaLM-E/PaLI-3; SigLIP + SoViT; Gemma-2B; CLIP → Flamingo/BLIP-2; Pix2Seq/LocCa/SAM; Prefix-LM; big_vision.
- Subsequent (clear): π_0 (PaliGemma init); $\pi_{0.5}$ / OpenPI; BEHAVIOR’25 1st (2512.06951).
- Arrow: SigLIP → **PaliGemma** → π_0 → $\pi_{0.5}$ /OpenPI → BEHAVIOR’25.

Visionary (11+3) — seat 5

2026 BEHAVIOR: PaliGemma-class remains default open VLA init until a better open egocentric/video 3–4B appears. Upside = **adaptation** (action expert, noise, stages, data), not backbone replacement. Multi-cam 224 + selective 448 foveation > blanket 896. Winners already swap free language for task/stage embeddings — treat PG as visual tokenizer + residual LLM. Skip Discord/Drive.

4. VLA / control note

Perception/grounding backbone (loc/seg, OCR@448+), not a policy or classical planner. Prefix-LM fits observe+goal → action shaping; don't freeze vision early (slow warmup). Open weights are load-bearing for BEHAVIOR forks.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Open 3B substrate → robot FMs
Reviewer	**8+3**	**Strengths / weaknesses / 10 actionable asks (HIGH)**
Empiricist	11+3	Res ladder + LoRA freeze matrix
Archaeologist	11+3	SigLIP → PG → π_0 → BEHAVIOR timeline
Visionary	11+3	2026 adaptation > backbone swap

Image-first; speaker notes; **PDF for Mac**.

Paths

Artifact	Path
This pack	`/workspace/paligemma-quiet/KANTA_PACK.md`
Senior Research	`/workspace/paligemma-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/paligemma-2407.07726-summary.md`